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# Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

### Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssword” | base64
```

### Step 2: Create a Secret from a File

```
C:\Users\prati>kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-04-14T13:07:07Z"
  name: my-secret
  namespace: default
  resourceVersion: "7049"
  uid: 06ae8cf0-0a36-4ff9-9a41-dff8bc15b5d0
type: Opaque
```

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt



echo -n 'my-password' > ./password.txt
```

```
C:\Users\prati>echo my-username > username.txt

C:\Users\prati>echo -n 'my-password' > ./password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

```
C:\Users\prati>kubectl create secret generic my-file-secret --from-file=username=username.txt --from-file=password=passw
ord.txt
secret/my-file-secret created
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
C:\Users\prati>kubectl get secret my-file-secret -o yaml
apiVersion: v1
data:
  password: LW4gJ215LXBhc3N3b3JkJyANCg==
  username: bXktdXNlcm5hbWUgDQo=
kind: Secret
metadata:
  creationTimestamp: "2026-04-14T13:10:56Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "7233"
  uid: 1bfd0768-1523-47c0-8684-22d6d86aa8a3
type: Opaque
```

**Step 3: Create a Docker Registry Secret**

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

```
C:\Users\prati>kubectl create secret docker-registry my-registry-secret --docker-username=your_username --docker-passwor
d=your_password --docker-email=your_email --docker-server=docker.io
secret/my-registry-secret created
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

```
C:\Users\prati>kubectl get secret my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJkb2NrZXIuaW8iOnsidXNlcm5hbWUiOiJ5b3VyX3VzZXJuYW1lIiwicGFzc3dvcmQiOiJ5b3VyX3Bhc3N3b3J
kIiwiZW1haWwiOiJ5b3VyX2VtYWlsIiwiYXV0aCI6ImVXOTFjbDkxYzJWeWJtRnRaVHA1YjNWeVgzQmhjM04zYjNKayJ9fX0=
kind: Secret
metadata:
  creationTimestamp: "2026-04-14T13:11:24Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "7257"
  uid: 2b73791b-5b44-4aba-a315-bc3c455324c2
type: kubernetes.io/dockerconfigjson
```

**Step 4: Create a Secret Using a YAML Manifest**

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
C:\Users\prati>notepad my-manifest-secret.yaml

C:\Users\prati>kubectl apply -f my-manifest-secret.yaml
secret/my-manifest-secret created
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
C:\Users\prati>kubectl get secret my-manifest-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","data":{"password":"UEBzc3cwcmQ=","username":"YWRtaW4="},"kind":"Secret","metadata":{"annotatio
ns":{},"name":"my-manifest-secret","namespace":"default"},"type":"Opaque"}
  creationTimestamp: "2026-04-14T13:12:07Z"
  name: my-manifest-secret
  namespace: default
  resourceVersion: "7292"
  uid: 93d54f11-ca65-4daa-ae6d-bb1c152193f0
type: Opaque
```

### Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

```
C:\Users\prati>kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
```

```
C:\Users\prati>kubectl get pods
NAME                          READY   STATUS              RESTARTS        AGE
devops-app-6fc4b9fb96-999vw   1/1     Running             4 (8m15s ago)   3d17h
devops-app-6fc4b9fb96-p9tb4   1/1     Running             4 (8m15s ago)   3d17h
pod-with-secret               0/1     ContainerCreating   0               6s
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

**Step 7: Clean Up**

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
C:\Users\prati>kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted from default namespace

C:\Users\prati>kubectl delete secret my-secret
secret "my-secret" deleted from default namespace

C:\Users\prati>kubectl delete pod pod-with-secret
Error from server (NotFound): pods "pod-with-secret" not found

C:\Users\prati>kubectl delete secret my-file-secret
secret "my-file-secret" deleted from default namespace

C:\Users\prati>kubectl delete secret my-registry-secret
secret "my-registry-secret" deleted from default namespace

C:\Users\prati>kubectl delete secret my-manifest-secret
secret "my-manifest-secret" deleted from default namespace
```